Differential Equations: Complete Tutorial with Examples and Practice

Calculus Tutorial

Contents

1 Introduction to Differential Equations

1.1 What is a Differential Equation?

A **differential equation** (DE) is an equation that relates a function with its derivatives. In simpler terms:

$$\boxed{\textbf{It describes how a quantity changes, and we need to find the quantity itself.}}$$

1.2 Basic Examples

- $\frac{dy}{dx} = 3x^2$ - First order DE (only first derivative)
- $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y = 0$ - Second order DE
- $x\frac{dy}{dx} + y = e^x$ - Linear DE

1.3 Why Study Differential Equations?

Differential equations model real-world phenomena:

- Population growth
- Radioactive decay
- Cooling/heating
- Motion under gravity
- Electrical circuits
- Chemical reactions

2 Basic Terminology

2.1 Order and Degree

- **Order:** The highest derivative in the equation
  Example: $\frac{d^3y}{dx^3} + 2\frac{dy}{dx} = 0$ has order 3
- **Degree:** The power of the highest derivative (when the equation is polynomial in derivatives)
  Example: $\left(\frac{dy}{dx}\right)^2 + y = 0$ has degree 2

2.2 General vs Particular Solution

- **General Solution:** Contains arbitrary constants
- **Particular Solution:** Constants are determined by initial conditions

3 Separable Differential Equations

3.1 What are Separable DEs?

A DE is separable if it can be written as:

$$\frac{dy}{dx} = f(x)g(y)$$

or equivalently:

$$h(y)dy = k(x)dx$$

3.2 Solution Method

1. Separate variables: All y terms with dy, all x terms with dx
2. Integrate both sides
3. Solve for y if possible

3.3 Example 1: Basic Separable DE

$$\frac{dy}{dx} = xy$$

Solution:

$$\begin{aligned}
\frac{dy}{dx} &= xy \\
\frac{1}{y}dy &= xdx \quad \text{(Separate variables)} \\
\int \frac{1}{y}dy &= \int xdx \quad \text{(Integrate both sides)} \\
\ln|y| &= \frac{x^2}{2} + C_1 \quad \text{(General solution in implicit form)} \\
y &= e^{x^2/2+C_1} = Ce^{x^2/2} \quad \text{(Explicit form)}
\end{aligned}$$

where $C = e^{C_1}$.

3.4 Example 2: Another Separable DE

$$\frac{dy}{dx} = \frac{x}{y}$$

Solution:

$$\begin{aligned}
\frac{dy}{dx} &= \frac{x}{y} \\
ydy &= xdx \quad \text{(Separate)} \\
\int ydy &= \int xdx \quad \text{(Integrate)} \\
\frac{y^2}{2} &= \frac{x^2}{2} + C \\
y^2 &= x^2 + 2C \\
y &= \pm\sqrt{x^2 + C} \quad \text{(where } C \text{ is redefined as } 2C\text{)}
\end{aligned}$$

4 Linear First Order Differential Equations

4.1 Standard Form

A linear first order DE has the form:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

4.2 Solving Using Integrating Factor

The **integrating factor** (IF) is:

$$\mu(x) = e^{\int P(x)dx}$$

Then the solution is:

$$y = \frac{1}{\mu(x)} \int \mu(x)Q(x)dx + \frac{C}{\mu(x)}$$

4.3 Example 1: Simple Linear DE

$$\frac{dy}{dx} + y = e^x$$

Solution:

$$\begin{aligned}
P(x) &= 1, \quad Q(x) = e^x \\
\text{Integrating Factor: } \mu(x) &= e^{\int 1dx} = e^x \\
\text{Multiply both sides by } \mu(x): \\
e^x \frac{dy}{dx} + e^x y &= e^{2x} \\
\frac{d}{dx}(ye^x) &= e^{2x} \quad \text{(Left side is derivative of } ye^x\text{)} \\
\int d(ye^x) &= \int e^{2x} dx \\
ye^x &= \frac{1}{2}e^{2x} + C \\
y &= \frac{1}{2}e^x + Ce^{-x}
\end{aligned}$$

4.4 Example 2: Linear DE with x coefficient

$$\frac{dy}{dx} + 2xy = x$$

Solution:

$$\begin{aligned}
P(x) &= 2x, \quad Q(x) = x \\
\mu(x) &= e^{\int 2x dx} = e^{x^2} \\
e^{x^2}\frac{dy}{dx} + 2xe^{x^2}y &= xe^{x^2} \\
\frac{d}{dx}(ye^{x^2}) &= xe^{x^2} \\
ye^{x^2} &= \int xe^{x^2}\,dx \\
\text{Let } u = x^2, \quad du = 2xdx: & \\
\int xe^{x^2}dx &= \frac{1}{2}\int e^u du = \frac{1}{2}e^{x^2} \\
ye^{x^2} &= \frac{1}{2}e^{x^2} + C \\
y &= \frac{1}{2} + Ce^{-x^2}
\end{aligned}$$

5 Exact Differential Equations

5.1 Definition

A DE of the form:

$$M(x,y)dx + N(x,y)dy = 0$$

is **exact** if:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

5.2 Solution Method

If exact, there exists a function $F(x,y)$ such that:

$$\frac{\partial F}{\partial x} = M \quad \text{and} \quad \frac{\partial F}{\partial y} = N$$

and the solution is:

$$F(x,y) = C$$

5.3 Example: Exact DE

$$(2xy+1)dx + x^2 dy = 0$$

Solution:

$$\begin{aligned}
M &= 2xy+1, \quad N = x^2 \\
\frac{\partial M}{\partial y} &= 2x, \quad \frac{\partial N}{\partial x} = 2x \\
\text{Since } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}&\text{, the DE is exact} \\
\text{Find } F(x,y) \text{ such that:} & \\
\frac{\partial F}{\partial x} &= 2xy + 1 \quad (1) \\
\frac{\partial F}{\partial y} &= x^2 \quad (2)
\end{aligned}$$

From (1):

$$F = \int (2xy + 1)dx = x^2y + x + g(y)$$

From (2):

$$\frac{\partial F}{\partial y} = x^2 + g'(y) = x^2 \Rightarrow g'(y) = 0 \Rightarrow g(y) = C_1$$

Thus:

$$F(x, y) = x^2y + x = C \quad \text{(Solution)}$$

6 Initial Value Problems (IVP)

6.1 What is an IVP?

An IVP consists of:

1. A differential equation
2. An initial condition (e.g., $y(x_0) = y_0$)

6.2 Example: IVP with Separable DE

$$\frac{dy}{dx} = 2x, \quad y(0) = 3$$

Solution:

$$\begin{aligned}
\frac{dy}{dx} &= 2x \\
dy &= 2xdx \\
\int dy &= \int 2xdx \\
y &= x^2 + C \quad \text{(General solution)} \\
\text{Apply } y(0) = 3 : & \\
3 &= 0^2 + C \Rightarrow C = 3 \\
y &= x^2 + 3 \quad \text{(Particular solution)}
\end{aligned}$$

6.3 Example: IVP with Linear DE

$$\frac{dy}{dx} + y = e^x, \quad y(0) = 2$$

Solution: From earlier example, general solution is:

$$y = \frac{1}{2}e^x + Ce^{-x}$$

Apply $y(0) = 2$:

$$\begin{aligned}
2 &= \frac{1}{2}e^0 + Ce^0 \\
2 &= \frac{1}{2} + C \\
C &= \frac{3}{2}
\end{aligned}$$

Thus:

$$y = \frac{1}{2}e^x + \frac{3}{2}e^{-x}$$

7 Applications of Differential Equations

7.1 Exponential Growth and Decay

General form:

$$\frac{dP}{dt} = kP$$

where $k > 0$ for growth, $k < 0$ for decay.

Solution: $P(t) = P_0 e^{kt}$

7.2 Newton's Law of Cooling

$$\frac{dT}{dt} = -k(T - T_s)$$

where T is object temperature, T_s is surrounding temperature.

Solution: $T(t) = T_s + (T_0 - T_s)e^{-kt}$

7.3 Simple Harmonic Motion

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

Solution: $x(t) = A\cos(\omega t) + B\sin(\omega t)$

8 15 Practice Problems with Solutions

8.1 Separable Differential Equations

1. $\frac{dy}{dx} = y$

   Solution:

$$\begin{aligned}
\frac{dy}{dx} &= y \\
\frac{1}{y}dy &= dx \\
\int \frac{1}{y}dy &= \int dx \\
\ln|y| &= x + C \\
y &= e^{x+C} = Ce^x \quad (C > 0)
\end{aligned}$$

2. $\frac{dy}{dx} = x^2$

   Solution:

$$\begin{aligned}
\frac{dy}{dx} &= x^2 \\
dy &= x^2 dx \\
\int dy &= \int x^2 dx \\
y &= \frac{x^3}{3} + C
\end{aligned}$$

3. $\dfrac{dy}{dx} = xy^2$
Solution:

$$\begin{aligned}
\frac{dy}{dx} &= xy^2 \\
\frac{1}{y^2}dy &= xdx \\
\int y^{-2}dy &= \int xdx \\
-\frac{1}{y} &= \frac{x^2}{2} + C \\
y &= -\frac{1}{\frac{x^2}{2} + C} = -\frac{2}{x^2 + 2C}
\end{aligned}$$

4. $\dfrac{dy}{dx} = \dfrac{1}{y}$
Solution:

$$\begin{aligned}
\frac{dy}{dx} &= \frac{1}{y} \\
ydy &= dx \\
\int ydy &= \int dx \\
\frac{y^2}{2} &= x + C \\
y^2 &= 2x + 2C = 2x + C' \quad (C' = 2C)
\end{aligned}$$

8.2 Linear Differential Equations

5. $\dfrac{dy}{dx} + y = 0$
Solution:

$$\begin{aligned}
\frac{dy}{dx} + y &= 0 \\
P(x) = 1&, \quad Q(x) = 0 \\
\mu(x) &= e^{\int 1dx} = e^x \\
e^x\frac{dy}{dx} + e^x y &= 0 \\
\frac{d}{dx}(ye^x) &= 0 \\
ye^x &= C \\
y &= Ce^{-x}
\end{aligned}$$

6. $\dfrac{dy}{dx} + 2y = e^x$

Solution:

$$\begin{aligned}
P(x) &= 2, \quad Q(x) = e^x \\
\mu(x) &= e^{\int 2dx} = e^{2x} \\
e^{2x}\frac{dy}{dx} + 2e^{2x}y &= e^{3x} \\
\frac{d}{dx}(ye^{2x}) &= e^{3x} \\
ye^{2x} &= \frac{1}{3}e^{3x} + C \\
y &= \frac{1}{3}e^{x} + Ce^{-2x}
\end{aligned}$$

7. $\dfrac{dy}{dx} - y = x$

   Solution:

$$\begin{aligned}
\frac{dy}{dx} - y = x \quad &\Rightarrow \quad \frac{dy}{dx} + (-1)y = x \\
P(x) = -1, \quad &Q(x) = x \\
\mu(x) = e^{\int(-1)dx} &= e^{-x} \\
e^{-x}\frac{dy}{dx} - e^{-x}y &= xe^{-x} \\
\frac{d}{dx}(ye^{-x}) &= xe^{-x}
\end{aligned}$$

   Using integration by parts for $\int xe^{-x}dx$:

$$\begin{aligned}
\int xe^{-x}dx &= -xe^{-x} - \int(-e^{-x})dx \\
&= -xe^{-x} - e^{-x} + C \\
ye^{-x} &= -xe^{-x} - e^{-x} + C \\
y &= -x - 1 + Ce^{x}
\end{aligned}$$

8.3 Exact Differential Equations

8. $(y + 2x)dx + xdy = 0$

   Solution:

$$\begin{aligned}
M &= y + 2x, \quad N = x \\
\frac{\partial M}{\partial y} &= 1, \quad \frac{\partial N}{\partial x} = 1 \\
&\text{DE is exact} \\
\frac{\partial F}{\partial x} &= y + 2x \quad \Rightarrow \quad F = xy + x^2 + g(y) \\
\frac{\partial F}{\partial y} &= x + g'(y) = x \quad \Rightarrow \quad g'(y) = 0 \Rightarrow g(y) = C_1 \\
F(x, y) &= xy + x^2 = C
\end{aligned}$$

9. $(3x^2y)dx + (x^3)dy = 0$

Solution:

$$\begin{aligned}
M = 3x^2y, &\quad N = x^3 \\
\frac{\partial M}{\partial y} = 3x^2, &\quad \frac{\partial N}{\partial x} = 3x^2
\end{aligned}$$

DE is exact

$$\begin{aligned}
\frac{\partial F}{\partial x} &= 3x^2y \quad \Rightarrow \quad F = x^3y + g(y) \\
\frac{\partial F}{\partial y} &= x^3 + g'(y) = x^3 \quad \Rightarrow \quad g'(y) = 0 \Rightarrow g(y) = C_1 \\
F(x, y) &= x^3y = C
\end{aligned}$$

8.4 Initial Value Problems

10. $\dfrac{dy}{dx} = 4x, \quad y(1) = 5$

Solution:

$$\begin{aligned}
\frac{dy}{dx} &= 4x \\
dy &= 4xdx \\
\int dy &= 4\int xdx \\
y &= 2x^2 + C \\
y(1) = 5: \quad 5 &= 2(1)^2 + C \Rightarrow C = 3 \\
y &= 2x^2 + 3
\end{aligned}$$

11. $\dfrac{dy}{dx} = \sin x, \quad y(0) = 1$

Solution:

$$\begin{aligned}
\frac{dy}{dx} &= \sin x \\
dy &= \sin x dx \\
\int dy &= \int \sin x dx \\
y &= -\cos x + C \\
y(0) = 1: \quad 1 &= -\cos 0 + C \Rightarrow 1 = -1 + C \Rightarrow C = 2 \\
y &= -\cos x + 2
\end{aligned}$$

8.5 Mixed Problems

12. $\dfrac{dy}{dx} = e^{2x}$

Solution:

$$\begin{aligned}
\frac{dy}{dx} &= e^{2x} \\
dy &= e^{2x}dx \\
\int dy &= \int e^{2x}dx \\
y &= \frac{1}{2}e^{2x} + C
\end{aligned}$$

13. $\dfrac{dy}{dx} + xy = 0$
Solution:

$$\begin{aligned}
\frac{dy}{dx} + xy &= 0 \\
\frac{dy}{dx} &= -xy \quad \text{(This is separable)} \\
\frac{1}{y} dy &= -x dx \\
\int \frac{1}{y} dy &= -\int x dx \\
\ln|y| &= -\frac{x^2}{2} + C \\
y &= e^{-x^2/2+C} = Ce^{-x^2/2}
\end{aligned}$$

14. $\dfrac{dy}{dx} = y \ln y$
Solution:

$$\begin{aligned}
\frac{dy}{dx} &= y \ln y \\
\frac{1}{y \ln y} dy &= dx \\
\int \frac{1}{y \ln y} dy &= \int dx \\
\text{Let } u = \ln y, \quad du &= \frac{1}{y} dy : \\
\int \frac{1}{u} du &= x + C \\
\ln|u| &= x + C \\
\ln|\ln y| &= x + C \\
\ln y &= e^{x+C} = Ce^{x} \quad (C > 0) \\
y &= e^{Ce^{x}}
\end{aligned}$$

9 Exam Tips and Strategies

9.1 How to Approach DE Problems

1. **Identify the type:** Check if separable, linear, exact, etc.
2. **Choose appropriate method:**
   - Variables can be separated? $\rightarrow$ Separable method
   - Can be written as $\frac{dy}{dx} + P(x)y = Q(x)$? $\rightarrow$ Integrating factor
   - $Mdx + Ndy = 0$ with $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$? $\rightarrow$ Exact DE
3. **Solve step by step:** Don't skip integration steps
4. **Check initial conditions:** If given, find particular solution
5. **Verify:** Differentiate your solution to check it satisfies the DE

9.2 Common Mistakes to Avoid

1. **Forgetting the constant of integration** for indefinite integrals
2. **Misidentifying the type** of differential equation
3. **Algebraic errors** when separating variables
4. **Incorrect integrating factor** calculation
5. **Not applying initial conditions** when finding particular solutions

9.3 Quick Reference Table

Type	**Form**	**Method**
Separable	$\frac{dy}{dx} = f(x)g(y)$	Separate variables and integrate
Linear 1st order	$\frac{dy}{dx} + P(x)y = Q(x)$	Integrating factor: $\mu = e^{\int P dx}$
Exact	$Mdx + Ndy = 0$, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$	Find F such that $dF = Mdx + Ndy$
Homogeneous	$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$	Substitute $v = \frac{y}{x}$

10 Practice for Mastery

To master differential equations:

1. Practice identifying types quickly
2. Work through examples without looking at solutions
3. Create your own problems and solve them
4. Focus on applications to understand real-world meaning
5. Time yourself to build exam speed

10.1 Final Exam Tip

Always check: Can I separate variables? If not, is it linear? If not, is it exact?